Kenneth Lieu

415-658-1852| kennethllieu3@gmail.com | Portfolio | LinkedIn | GitHub | Twitter

Education

University of California, Davis Davis, CA

Bachelor of Science in Computer Science and Engineering, Minor in Electrical Engineering *Sep. 2018 – June 2022*

Experience

Semiconductor Product Validation Technician (contract) April 2026 - October 2026

NVIDIA *Santa Clara, CA*

- **Hardware Integration and Assembly:** Assembled high-density server boards with 20+ specialized components (GPU, BMC, MCU), adhering strictly to mechanical specifications.
- **Thermal Validation & Bench Testing:** Executed thermal characterization and stress testing across GPU test suites, validating system stability and I2C telemetry metrics under load.
- **Precision Bench Maintenance:** Applied liquid metal thermal interface materials (TIM) and replaced precision thermal pads on complex board assemblies, adhering to strict component tolerances.
- **Diagnostics & Ticket Execution:** Isolated root causes for 30+ complex issues and job-cancellation errors; resolved 100+ tickets, cutting resolution time by 15%.

Debug Repair Technician (contract) Feb 2026 – April 2026

Quanta Manufacturing *Fremont, CA*

- **Failure Analysis & Debugging:** Performed root-cause analysis on server boards to isolate and repair 20+ hardware defects; authored 20 Shell scripts in Linux/Cygwin to automate feature debugging.
- **Reporting & Metrics Tracking:** Summarized repair diagnostics across 3 internal applications and Excel to track defect metrics and improve board yield.

Thermal SQA Engineer (contract) Jan 2025 – May 2025

Apple *Cupertino, CA*

- **Thermal characterization & Automation:** Executed Python and Linux automation scripts across 10 Thermal SQA test suites to validate heat dissipation on Mac prototype hardware.
- **Instrumentation & Assembly:** Instrumented Mac assemblies with thermocouples to quantify component thermal profiles across 3 controlled environmental conditions using precision hand tools.

Server Repair Technician (contract) Nov. 2023 – Mar. 2024

Wistron Corporation *Santa Clara, CA*

- **Board-Level Failure Analysis:** Executed failure analysis on Vulcan and Viking server boards to isolate digital circuitry defects; built automated Linux CLI quality assurance tests.
- **Hardware Assembly:** Assembled 9 core board components to streamline hardware diagnostic procedures and support functional testing.

Projects

Clear Lake Data Visualizations | *JavaScript, React, HTML/CSS, SQL* Jan. 2022 – Jun. 2022

- **Frontend Architecture:** Architected a dynamic React web application using JavaScript, HTML, and CSS to plot interactive weather metrics, wind maps, and custom data filters.
- **Data Integration:** Integrated external data packages and relational SQL queries to process and render real-time environmental data visualizations.

UNO Lightning | *C++, Linux* Jun. 2023 – Present

- **Game Engine Mechanics:** Engineered an expanded C++ card game engine in Linux implementing custom rules (LIGHTNING and PARDON) utilizing Object-Oriented Programming.
- **Code Optimization & Testing:** Applied modular class abstraction to reduce code complexity and executed systematic unit testing to validate turn-based logic constraints.

Technical Skills

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS, MATLAB, Rust, Go
Hardware & Validation: Hardware Troubleshooting, Failure Analysis, Thermal Validation, TIM & Liquid Metal Application, Vulcan Boards, Oscilloscopes, Voltmeters, Multimeters, ADALM2000, I2C Telemetry
Frameworks & Platforms: React, Node.js, JUnit, AWS, Linux, Cygwin, Shell Scripting
Developer Tools: Git, GitHub, BitBucket, Docker, VS Code, PyCharm, Eclipse, SourceTree